

1 Introduction

During this sprint we visited the Bosch offices in Belgrade and collected the platform. After returning home we began performing maintenance on the car while also continuing development on the codebase. We took some time off to celebrate the holidays with our families and then got right back to work. We had some setbacks, like getting kicked out from our previous workspace but we managed to find a new one and continued work as planned.

2 Planned activities

During this sprint, we had a plan to accomplish the following tasks:

- Traffic sign detection and line detection in simulation
- Deploy code on Nucleo and RPi, and set everything up
- Collect images from simulation and additional model training
- Collect images from the car camera
- Making a physical environment
- Completely disassemble and reassemble the car for maintenance
- Further development of middleware

3 Status of planned activities

- **Traffic sign detection and line detection in simulation**
We were advised that we should focus working on the actual platform over working in the simulation, so we shifted our work to the physical platform in a real environment.
- **Deploy code on Nucleo and RPi, and set everything up**
We got on this task as soon as we brought the platform home and got it done without major issues.
- **Collect images from simulation and additional model training**
We decided to postpone this task and focus on implementing other systems before fully committing our attention to sign detection.
- **Collect images from the car camera**
As previously mentioned, we decided that sign detection wasn't a priority yet, however, we did deploy a pre-trained YOLO model to see how much strain that would put on the system. We intend to train a model on our own dataset in the next sprint.
- **Making a physical environment**
We were lucky to find out that a model of the challenge circuit was donated to our faculty a few years ago and have since started using it to test our work.
- **Completely disassemble and reassemble the car for maintenance**
Having picked up the car we began taking it apart to figure out how everything was set up, how it worked and to find out if anything needed fixing due to shipping. We regreased the differentials, replaced many stripped or missing screws, fixed the servo to the steering rack, fixed an issue where the steering rack would rub against the driveshaft and adjusted the wheels. We will likely do this again in one of the later sprints, just in case.
- **Further development of middleware**
We finished developing the middleware. We made Docker containers for the simulator and the brain. We use it when we need to test software in parallel, however it is very rare that we need to test multiple codebases at the same time and because of that the middleware is currently not being used most of the time.

4 General status of the project

The car can drive itself and follow a lane. In the last report, we had a line detection algorithm that took points out of an image and drew contours to connect them, giving us our lines. In this split we tried to implement a control algorithm using those points which didn't work as well as we wanted. Instead, we switched to using canny algorithm and making a distinction between the left and right lane. We then use the angle of the road and compare it to where the car is pointed to get an input for a PID regulator which turns the wheels to follow the road.

The car also considers twists and turns in the road and adjusts its own speed to take turns as smoothly as possible. It speeds up when the road is straight and slows down to follow turns.

We are in the process of implementing sign detection and implementing a way for the car to know how to react to intersections properly. For now, the car stops when it detects an intersection.

We also set up a flask server to show us a multiplexed image of multiple parts of the project (raw camera output, lane detection output, object (sign) detection output) which we use as a debugging tool. It works well but it does have latency issues which we intend to fix by making the implementation as simple as possible (probably with UDP).

As for now, our biggest issue is figuring out how to fix latency issues caused by the processor being overloaded by the pretrained YOLO model. We tried to rectify this by limiting the number of threads the YOLO model can create. We are currently reading up on the subject and expect to fix this issue when we train a model on our own dataset.

5 Upcoming activities

In the next sprint we want to do the following tasks:

- Fix latency issues
- Collect images from the car camera
- Traffic sign detection
- Intersection traversal
- Begin working on path planning
- Possible minor hardware improvements